

Practical Exercise

Google Big Query Dataset: Retail Sales Dataset

QUESTIONS:

Q1 WHERE Clause

The screenshot shows the Google BigQuery console interface. The left sidebar contains navigation options: Overview, Studio (selected), Agents, Pipelines and integration, Data transfers, Dataform, Scheduled queries, Scheduling, Governance, Sharing (Analytics Hub), Policy tags, Metadata curation, Administration, Monitoring, Partner Centre, Settings, and Release notes. The main area displays 'Question 1' with a SQL query:

```
1 SELECT *
2 FROM `sql-big-query-494420.retail_data.retail_sales`
3 WHERE EXTRACT(YEAR FROM Date) = 2023;
4
5
```

Below the query, a status message indicates: 'This query will process 72.69 KB when run. Using on-demand processing quota'. The 'Query results' section shows a table with 10 rows of data. The table has columns: Row, Transaction ID, Date, Customer ID, Gender, Age, Product Category, and Quantity. The results are as follows:

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity
1	191	2023-10-18	CUST1191	Male	64	Beauty	
2	204	2023-09-28	CUST204	Male	39	Beauty	
3	230	2023-04-23	CUST230	Male	54	Beauty	
4	232	2023-02-06	CUST232	Female	43	Beauty	
5	309	2023-12-23	CUST309	Female	26	Beauty	
6	310	2023-10-12	CUST310	Female	28	Beauty	
7	363	2023-06-03	CUST363	Male	64	Beauty	
8	371	2023-02-21	CUST371	Female	20	Beauty	
9	397	2023-03-10	CUST397	Female	30	Beauty	
10	454	2023-02-22	CUST454	Female	46	Beauty	

At the bottom, the 'Job history' section is visible, showing 'Results per page: 50' and '1 - 50 of 998'.

Q2 Filtering + Conditions

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains the Explorer panel with a search bar and a list of resources for the project 'sql-big-query-494420', including Datasets, Connections, Conversations, Agents, Queries, (Classic) Queries, Notebooks, Data canvases, Data preparations, Pipelines, and Repositories. The main panel displays 'Question 2' with the following SQL query:

```
1 SELECT *
2 FROM `sql-big-query-494420.retail_data.retail_sales`
3 WHERE `Total Amount` > (
4   SELECT AVG(`Total Amount`)
5   FROM `sql-big-query-494420.retail_data.retail_sales`
6 );
```

Below the query, a status bar indicates 'Query completed' and 'Using on-demand processing quota'. The 'Query results' section shows a table with 9 rows of data. The table has columns: Row, Transaction ID, Date, Customer ID, Gender, Age, Product Category, and Quant. The data is as follows:

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quant
1	21	2023-01-14	CUST021	Female	50	Beauty	
2	28	2023-04-23	CUST028	Female	43	Beauty	
3	128	2023-07-05	CUST128	Male	25	Beauty	
4	220	2023-03-03	CUST220	Male	64	Beauty	
5	238	2023-01-17	CUST238	Female	39	Beauty	
6	364	2023-08-23	CUST364	Female	19	Beauty	
7	408	2023-04-15	CUST408	Female	64	Beauty	
8	537	2023-06-03	CUST537	Female	21	Beauty	
9	539	2023-06-08	CUST539	Male	25	Beauty	

At the bottom, the 'Job history' section is visible with a 'Show' link.

Q3 Aggregate Functions

The screenshot shows the Google Cloud BigQuery console interface for 'Question 3'. The left sidebar is identical to the previous screenshot. The main panel displays the following SQL query:

```
1 SELECT SUM(`Total Amount`) AS Total_Revenue
2 FROM `sql-big-query-494420.retail_data.retail_sales`;
3
```

Below the query, a status bar indicates 'This query will process 7.81 KB when run.' and 'Using on-demand processing quota'. The 'Query results' section shows a table with 1 row of data. The table has columns: Row and Total_Revenue. The data is as follows:

Row	Total_Revenue
1	456000

At the bottom, the 'Job history' section is visible with a 'Show' link.

Q4 DISTINCT

The screenshot shows the Google Cloud BigQuery console interface. The top navigation bar includes the Google Cloud logo and a search bar. The left sidebar contains the Explorer panel with a search bar and a list of resources. The main area displays 'Question 4' with a SQL query and its results.

Query 4

```
1 SELECT DISTINCT 'Product Category'
2 FROM `sql-big-query-494420.retail_data.retail_sales`;
3
```

This query will process 10.17 KB when run.
Using on-demand processing quota

Query results

Job information Results Visualisation JSON Execution details Execution graph

Row	Product Category
1	Beauty
2	Clothing
3	Electronics

Results per page: 50 1 - 3 of 3

Job history [Show](#)

Q5 GROUP BY

The screenshot shows the Google Cloud BigQuery console interface. The top navigation bar includes the Google Cloud logo and a search bar. The left sidebar contains the Explorer panel with a search bar and a list of resources. The main area displays 'Question 5' with a SQL query and its results.

Question 5

```
1 SELECT 'Product Category',
2 SUM(Quantity) AS Total_Quantity
3 FROM `sql-big-query-494420.retail_data.retail_sales`
4 GROUP BY 'Product Category';
5
```

This query will process 17.98 KB when run.
Using on-demand processing quota

Query results

Job information Results Visualisation JSON Execution details Execution graph

Row	Product Category	Total_Quantity
1	Beauty	771
2	Clothing	894
3	Electronics	849

Results per page: 50 1 - 3 of 3

Job history [Show](#)

Q6 CASE Statement

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains the Explorer panel with a search bar and navigation links for Home, Starred, Shared with me, Job history, and a project-specific section for 'sql-big-query-494420' including Datasets, Connections, Conversations, Agents, Queries, (Classic) Queries, Notebooks, Data canvases, Data preparations, Pipelines, and Repositories.

The main panel displays 'Question 6' with a SQL query:

```
1 SELECT 'Customer ID',  
2       Age,  
3       CASE  
4         WHEN Age < 30 THEN 'Youth'  
5         WHEN Age BETWEEN 30 AND 59 THEN 'Adult'  
6         ELSE 'Senior'  
7       END AS Age_Group  
8 FROM `sql-big-query-494420.retail_data.retail_sales`;  
9
```

Below the query, a status message indicates: "This query will process 16.6 KB when run."

The 'Query results' section shows a table with columns: Customer ID, Age, and Age_Group. The results are as follows:

Row	Customer ID	Age	Age_Group
1	CUST191	64	Senior
2	CUST204	39	Adult
3	CUST230	54	Adult
4	CUST232	43	Adult
5	CUST309	26	Youth
6	CUST310	28	Youth
7	CUST363	64	Senior
8	CUST371	20	Youth

At the bottom, the 'Job history' section is visible with a 'Show' link.

Q7 Conditional Aggregation

The screenshot shows the Google Cloud BigQuery console interface for 'Question 7'. The SQL query is:

```
1 SELECT Gender,  
2       COUNTIF('Total Amount' > 500) AS High_Value_Transactions  
3 FROM `sql-big-query-494420.retail_data.retail_sales`  
4 GROUP BY Gender;  
5
```

A status message indicates: "This query will process 14.67 KB when run." and "Using on-demand processing quota".

The 'Query results' section shows a table with columns: Gender and High_Value_Transactions. The results are as follows:

Row	Gender	High_Value_Transactions
1	Male	144
2	Female	155

At the bottom, the 'Job history' section is visible with a 'Show' link.

Q8 HAVING Clause

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains the Explorer panel with a search bar and navigation links for Home, Starred, Shared with me, Job history, and various data management tools. The main panel displays 'Question 8' with a SQL query that uses a HAVING clause to filter results where the total amount is greater than 5000. The query is as follows:

```
1 SELECT 'Product Category',
2       SUM('Total Amount') AS Total_Revenue
3 FROM `sql-big-query-494420.retail_data.retail_sales`
4 GROUP BY 'Product Category'
5 HAVING SUM('Total Amount') > 5000;
```

Below the query, a status message indicates that the query will process 17.98 KB when run, using on-demand processing quota. The 'Query results' section shows a table with three rows of data:

Row	Product Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

The interface also includes tabs for Job information, Results, Visualisation, JSON, Execution details, and Execution graph. The 'Results' tab is currently selected, showing the query results. The bottom of the interface shows the 'Job history' section with a 'Show' link.

Q9 Calculated Fields

The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains the Explorer panel with a search bar and navigation links for Home, Starred, Shared with me, Job history, and various data management tools. The main panel displays 'Question 9' with a SQL query that uses a CASE statement to categorize transactions based on their price per unit. The query is as follows:

```
1 SELECT 'Transaction ID',
2       'Price per Unit',
3       CASE
4         WHEN 'Price per Unit' < 50 THEN 'Cheap'
5         WHEN 'Price per Unit' BETWEEN 50 AND 200 THEN 'Moderate'
6         ELSE 'Expensive'
7       END AS Unit_Cost_Category
8 FROM `sql-big-query-494420.retail_data.retail_sales`;
```

Below the query, a status message indicates that the query will process 15.63 KB when run, using on-demand processing quota. The 'Query results' section shows a table with six rows of data:

Row	Transaction ID	Price per Unit	Unit_Cost_Category
1	191	25	Cheap
2	204	25	Cheap
3	230	25	Cheap
4	232	25	Cheap
5	309	25	Cheap
6	310	25	Cheap

The interface also includes tabs for Job information, Results, Visualisation, JSON, Execution details, and Execution graph. The 'Results' tab is currently selected, showing the query results. The bottom of the interface shows the 'Job history' section with a 'Show' link.

Q10 Combining WHERE + CASE

The screenshot shows the Google Cloud BigQuery console interface. The browser tabs include 'Anchor of Hope', 'big query - Google Search', 'Datasets - BigQuery - SQL B...', 'brightlearn portal - Google S...', and 'BrightLearn'. The address bar shows the BigQuery console URL. The left sidebar contains the 'Explorer' panel with a search bar and a list of resources: Home, Starred, Shared with me, Job history, and a project named 'sql-big-query-494420'. The main panel displays 'Question 10' with a SQL query and its results.

SQL Query:

```
1 SELECT 'Customer ID',
2       Age,
3       'Total Amount',
4       CASE
5         WHEN 'Total Amount' > 1000 THEN 'High'
6         ELSE 'Low'
7       END AS Spending_Level
8 FROM 'sql-big-query-494420.retail_data.retail_sales'
9 WHERE Age >= 40;
```

Query results:

Row	Customer ID	Age	Total Amount	Spending_Level
1	CUST191	64	25	Low
2	CUST230	54	25	Low
3	CUST232	43	25	Low
4	CUST363	64	25	Low
5	CUST454	46	25	Low
6	CUST512	57	25	Low
7	CUST791	51	25	Low

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